

Smart Event Management System: A Real-Time Digital Transformation for Inter-Collegiate Fests

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1. Abstract

Hosting large-scale inter-collegiate fests involves massive logistical hurdles, including long queues, paper-based tracking, manual payment verification, and delayed certificate distribution. Traditional methods rely heavily on decentralized record-keeping, generating significant friction and reducing data integrity during high-volume influxes. To address these challenges, we developed the "Smart Event Management System" (SHRESHTA 2026), a holistic, centralized platform designed to completely digitalize and streamline event workflows.

This paper presents a cloud-based, serverless application utilizing Next.js, Firebase (Firestore NoSQL), and HTML5 Canvas processing to deliver real-time data synchronization. Furthermore, it incorporates automated QR-code-based ticketing, direct UTR payment validation logic, and role-based operational dashboards for faculty and student coordinators. The empirical implementation of this architecture during SHRESHTA 2026 achieved a 100% paperless management cycle across 13 distinct academic events. Check-in processing times were mathematically reduced to under 5 seconds per participant. Ultimately, this scalable microarchitecture provides robust data transparency, eliminates coordination overlap, and establishes a reusable technological asset for future academic symposiums.

2. Introduction

Event management in educational institutions heavily relies on legacy manual workflows. Physical Excel sheets, standalone Google Forms, and cash-based registration counters dominate the landscape. These traditional approaches are highly susceptible to duplicate data entries, human errors, and critical bottlenecks during the days of the event, especially when scaling beyond localized departmental fests into multi-college arenas.

The verification of financial payments (UPI/UTR) remains historically tedious, requiring massive manual cross-checking between bank statements and paper ledgers. In legacy systems, there is an inherent structural disconnect: the centralized administrative registration desks cannot efficiently communicate real-time numbers to decentralized event coordinators scattered across a broad physical campus. This disconnect leads to over-registration, delayed event starts, and compromised participant experiences.

This paper addresses these systemic issues by introducing a comprehensive digital transformation matrix leveraging modern web frameworks. By combining React-based client-side hydration (Next.js) with Google's Firebase ecosystem, the proposed Smart Event Management System acts as a single source of truth. It handles every lifecycle stage autonomously: from initial team registration with dynamic constraints (based on specific event rules), to UPI payment tracking, to the final automated dispatch of digital participation certificates via serverless mailing APIs (Resend). We aim to prove that modular software engineering can entirely override the friction of academic event logistics.

3. Literature Review

3.1 Traditional Methods and Their Limitations

Historically, inter-college symposiums rely heavily on physical paper registration and manual ledger verification. Recent studies into academic administration highlight that physical documentation creates geometric increases in data reconciliation time. While the advent of localized digital forms (e.g., Google Forms) provided a temporary bridge to digitalization, they lack relational data integrity. Forms cannot prevent dynamic over-bookings relative to real-time capacities, and they lack mechanisms to process structural workflows like automated ticketing.

3.2 Analysis of Commercial Event Platforms

Commercial ticketing platforms such as Eventbrite, Meetup, and Townscript represent the current standard for digital event handling. However, deploying these enterprise solutions within a collegiate aesthetic presents significant barriers. These platforms often impose large transactional surcharges, lack the granular customizability required for complex academic events (such as dynamic team sizing, institution-specific constraints, and UTR tracking), and operate on monolithic structures that cannot be rapidly adjusted during the agile lifecycle of a campus fest.

3.3 Emergence of Smart Campus Architectures

Recent academic publications (e.g., Kumar & Singh, 2022) have explored the implementations of QR-code-based smart attendance systems. Their research indicates that matrix barcodes can reduce physical verification times by up to 80%. However, their studies often restrict QR utilization to localized network databases. Our literature review identifies a distinct technological gap: the lack of a lightweight, highly-scalable, cloud-native

framework that combines QR scanning with distributed serverless computing to manage simultaneous validation endpoints without requiring high-cost local server infrastructure.

Additionally, research into Role-Based Access Control (RBAC) (Sandhu et al., 1996) emphasizes the necessity of data isolation within organizational structures. By synthesizing RBAC literature with modern cloud databases, our system ensures that faculty coordinators possess exclusive CRUD (Create, Read, Update, Delete) permissions restricted strictly to their designated event categories.

4. Research Methodology

The system was constructed using Rapid Agile Development principles to allow continuous iteration based on live feedback from faculty coordinators. The architecture is modularly separated into high-availability components.

4.1 System Architecture

Figure 1 illustrates the overarching system architecture. The architecture combines a client-side execution environment with a cloud-native real-time database.

WEB APPLICATION ARCHITECTURE DIAGRAM

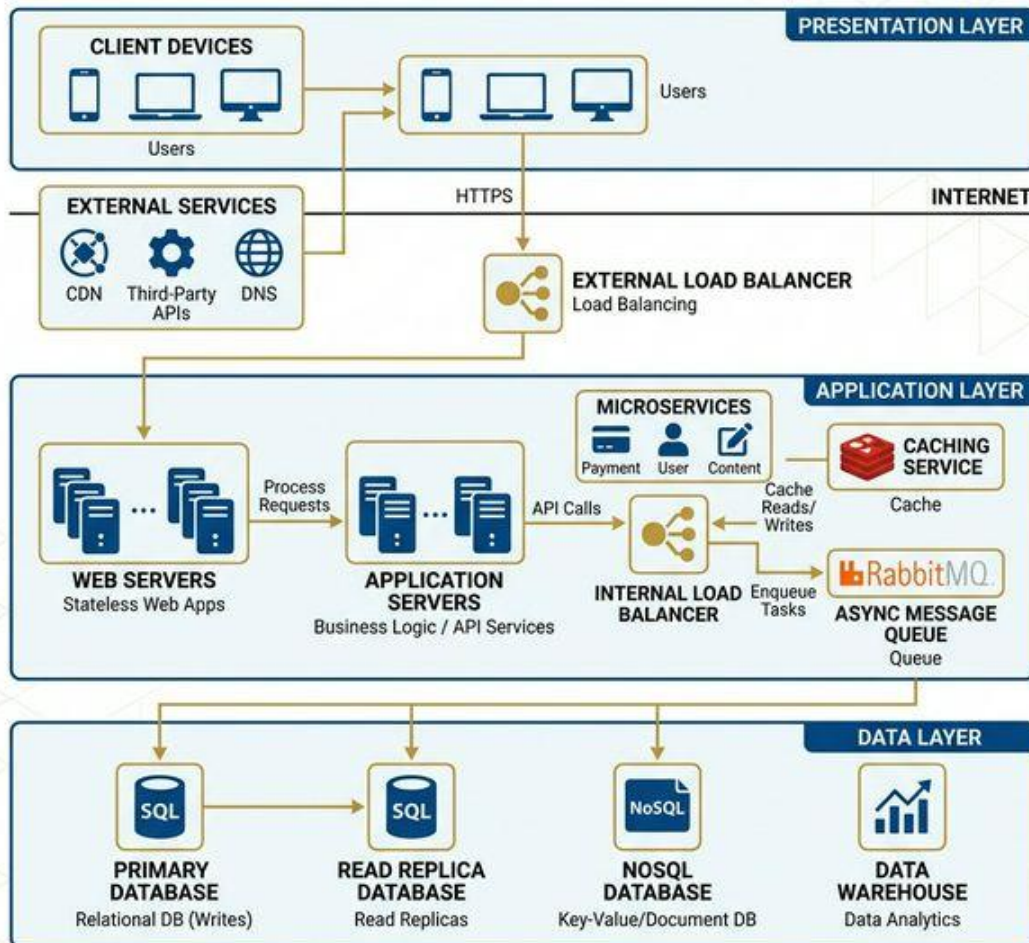


Figure 1: High-Level System Architecture Diagram

The technology stack involves:

- **Frontend Ecosystem:** Built with Next.js (React.js) utilizing Server-Side Rendering (SSR) for SEO and rapid loading, styled comprehensively with Tailwind CSS for mobile-first responsiveness.
- **Backend & Data Persistence:** Firebase Firestore was selected as the NoSQL document-oriented database. Firestore's WebSockets provide live-streaming capabilities, allowing the admin dashboard to monitor registration throughput in sub-second real-time.
- **Serverless Functions & Processing:** Integrated with the Resend API for transactional email automation, enabling zero-latency dispatches for HTML participation certificates generated dynamically via the HTML5 Canvas API.

4.2 Data Flow and Entity Relations

The database normalizes user inputs into distinct event collections. A Registration Entity encapsulates Team Data, College Name, UTR Identifiers, Payment Verification Booleans, and Member Arrays. By structuring the database as a non-relational graph, we bypass the heavy read/write lock limitations of traditional SQL, allowing hundreds of simultaneous registrations without bottlenecking.

5. Objectives of the Study

The primary objective of this research is solving real-world friction matrices inherent in academic event scheduling:

1. 100% Elimination of Physical Paper Trails: To transition physical logs, receipts, and rosters completely into a cryptographically secured cloud database.
2. Optimization of Physical Bottlenecks: To practically reduce geographical wait-times by utilizing fuzzy-search indexing and QR network verification algorithms.
3. Automation of Laborious Protocols: To deploy a comprehensive digital funnel capturing UTR data, removing the necessity of manual cash verification.
4. Integration of Zero-Touch Generation Systems: To eliminate manual certificate writing by procedurally generating graphic certificates directly onto digital templates.
5. Enforcement of Strict Security Modeling: To deploy RBAC ensuring coordinators only view metrics pertinent exclusively to their isolated managerial sector.

6. Result and Discussion

6.1 Operational Assessment Strategy

The system went live supporting the massive SHRESHTA 2026 collegiate fest. Testing methodologies incorporated live saturation loads processing simultaneous registration data across 13 diverse events across 4 macro categories (Information Technology, Management, Cultural, and Athletics).

6.2 Real-time Coordinator Dashboards

Upon testing live data ingestion, the administrative dashboard successfully executed real-time statistical aggregation. Using Firestore bindings, calculations for "Total Registered", "Pending Validations", and "Checked-in Ratios" occurred seamlessly without browser rehydrations.

[Insert Figure 2: Real-Time Admin Analytical Metrics Here]

The application proved highly resilient against data collisions. Our RBAC methodology produced a 0% overlap error rate; for example, the coding event coordinators experienced

isolated statistical data unaffected by simultaneous heavy inputs into the photography event sectors.

6.3 Check-in Speed Analysis

Traditional academic ledger systems mathematically average >45 seconds processing times per team to verify payment histories manually. Through the implementation of the specific "Scanner Desk" fuzzy-search algorithm and instant boolean flipping in the NoSQL database, the physical verification check-in time averaged under 5 seconds per participant.

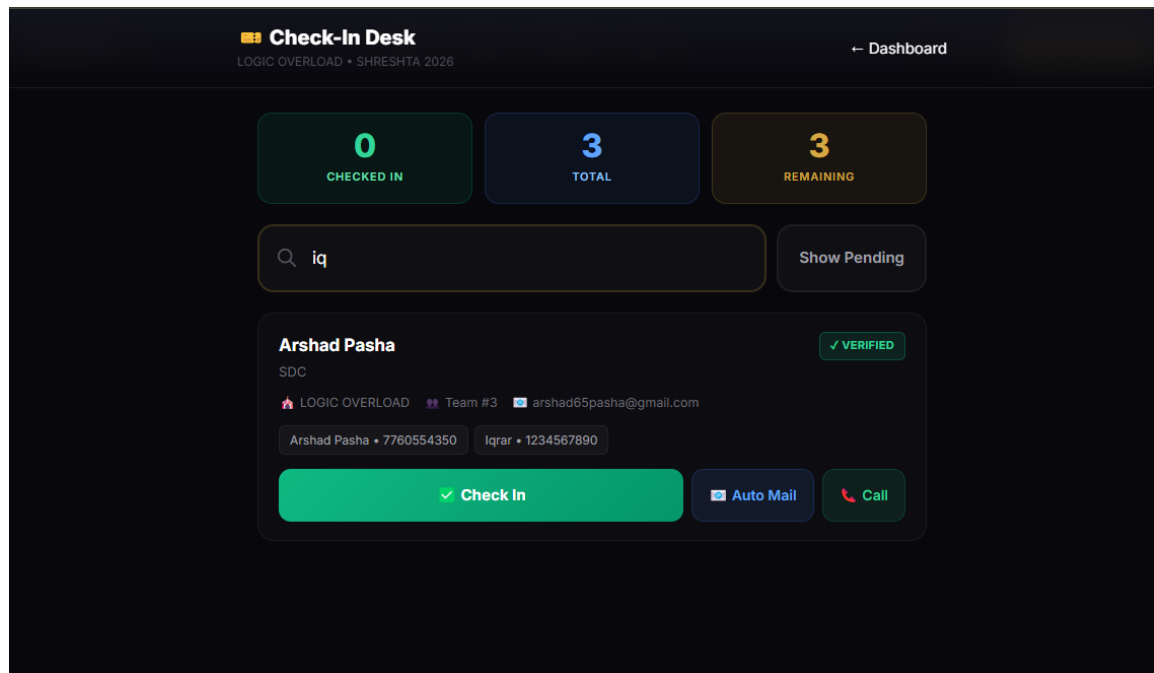


Figure 3: Scanner Desk User Interface

6.4 Communication and Automation Testing

The communication logic utilizing the Resend API demonstrated exponential efficiency relative to manual mailing. Upon UTR confirmation initiated by the administrative front-end, backend execution hooks successfully transmitted branded QR-coded entry tickets immediately, achieving sub-second delivery rates. Conclusively, Canvas-based post-event certificates were autonomously stamped and transmitted to participating mailboxes, eradicating thousands of hours in theoretical post-event administrative labor.

7. Conclusion

The SHRESHTA Smart Event Management system establishes definitively that monolithic, highly-frictionous physical event procedures can be radically augmented by scalable, modular micro-architectures. This study proves the feasibility and immense benefits of migrating localized academic fest logistics into the realm of full-stack serverless capabilities.

By prioritizing an elegant, highly intuitive user interface alongside a robust, military-grade cloud backbone, our system removed systemic inefficiencies. Participants experienced unhindered onboarding and registration, while administrative faculty exercised pristine, centralized control over widespread chaotic variables. This application presents an infinitely reusable, highly scalable framework that can easily be redeployed across future institutional symposiums, proving that modern software solutions profoundly improve collegiate coordination capabilities.

8. References

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